

SUPERIOR GRAND TRAVERSE BAY, MI, USA

WMO: 998176

Lat: 47.179N

Lon: 88.241W

Elev: 191

StdP: 99.05

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-19.3	N/A	N/A	N/A	N/A	N/A	N/A	11.2	-1.9	10.3	-2.1	3.0	320	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	28.2	N/A	26.3	N/A	24.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.1	7.9	-26.0	31.5	4.7	1.3	-29.4	32.4	-32.1	33.2	-34.8	33.9	-38.2	34.9		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.5	-8.6	-8.6	-4.1	2.3	7.3	12.8	17.7	17.5	13.6	7.2	1.2	-4.7
	DBStd	10.29	5.73	5.73	5.91	3.63	3.69	4.03	3.35	3.21	4.02	4.04	4.83	5.14
	HDD10.0	2728	576	522	436	234	103	16	0	0	12	107	264	457
	HDD18.3	5116	834	756	694	482	343	173	53	56	151	345	513	716
	CDD10.0	731	0	0	0	2	18	99	238	232	119	21	2	0
	CDD18.3	77	0	0	0	0	0	6	33	29	8	1	0	0
	CDH23.3	583	0	0	0	0	3	63	225	227	58	7	0	0
	CDH26.7	120	0	0	0	0	0	11	41	56	11	1	0	0
Wind	WSAvg	3.2	3.5	3.6	3.5	3.8	3.3	2.7	2.6	2.7	2.8	3.5	3.0	3.1
Precipitation	PrecAvg	678	40	30	30	51	72	73	72	56	85	73	51	42
	PrecMax	892	114	65	78	143	162	143	165	138	251	189	120	87
	PrecMin	484	1	11	0	14	33	15	27	9	31	25	16	7
	PrecStd	111	32	14	20	30	28	31	30	30	44	37	22	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	5.3	12.1	18.4	24.0	28.0	30.0	30.8	28.3	23.9	15.4	8.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	3.2	2.6	8.9	13.5	20.2	25.2	28.0	28.4	24.8	18.8	11.6	5.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	1.7	1.0	6.5	9.8	16.5	22.3	26.2	25.9	22.4	15.7	10.0	3.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	0.2	-0.2	4.1	7.5	13.8	19.3	24.3	23.8	20.4	13.5	8.5	2.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.9	8.7	10.4	8.0	8.6	9.1	10.1	10.3	10.3	7.8	6.6	6.3
		MCDBR	7.8	10.1	13.0	12.2	14.2	14.1	12.8	13.9	13.5	12.9	9.8	6.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.266	0.286	0.310	0.345	0.376	0.383	0.394	0.375	0.365	0.329	0.299	0.267	
	taud	2.441	2.364	2.373	2.362	2.372	2.370	2.391	2.442	2.455	2.521	2.522	2.476	
	Ebn at Noon	839	887	915	909	888	881	865	870	846	829	792	797	
	Edh at Noon	65	87	101	114	117	118	114	104	94	76	62	57	
All-Sky Solar Radiation	RadAvg	1.00	1.93	3.40	4.63	5.52	6.03	6.13	5.24	3.81	2.25	1.13	0.74	
	RadStd	0.09	0.23	0.34	0.52	0.39	0.42	0.28	0.35	0.21	0.27	0.17	0.09	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page